

US Patent & Trademark Office

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United States Patent Application Publication

20250279721

Kind Code

A1

Publication Date

September 04, 2025

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METHOD AND SYSTEM FOR BUFFERING A REFERENCE VOLTAGE

Abstract

In accordance with an embodiment, a circuit includes an amplifier comprising a first input coupled to a reference voltage node, a first buffer having an input coupled to an output of the amplifier, a second buffer having an input coupled to the output of the amplifier, and a capacitor. A method of operating the circuit includes during a first phase of operation: forming a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sampling a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, forming a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

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Family ID: 96738290

Appl. No.: 18/592754

Filed: March 01, 2024

Publication Classification

Int. Cl.: H02M3/155 (20060101); H03M1/06 (20060101)

U.S. Cl.:

CPC H02M3/155 (20130101); H03M1/06 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates generally to electronic systems and methods, and, in particular embodiments, to a method and system for a reference buffer.

BACKGROUND

[0002] Electronic systems often require the conversion of signals from analog to digital form and vice versa. Analog signals represent information in a continuous range of values, whereas digital signals represent information in discrete or quantized steps. Analog-to-digital converters (ADCs) are used to transform continuous analog signals into a digital representation, allowing for signal processing, storage, and transport in digital systems. Conversely, digital-to-analog converters (DACs) take digital data and reconstruct the original analog signal for use in applications such as audio playback, control systems, and communication devices.

[0003] Both ADCs and DACs are fundamental components in numerous electronic devices and systems, including computers, smartphones, medical equipment, instrumentation, and industrial control systems. To ensure accurate and reliable operation of these converters, reference voltages play an important role.

[0004] Reference voltages are used by ADCs and DACs as a benchmark to compare against input signals during the conversion process. The quality of these reference voltages may directly influence the accuracy and linearity of the conversion process. In some cases, offset and noise within the reference voltage may manifest as errors in the digital or reconstructed analog signal.

[0005] In ADCs, the reference voltage often defines the maximum input voltage range that can be digitized. Each step of digital value (quantization level) corresponds to a specific fraction of the reference voltage. Similarly, in DACs, the reference voltage may determine the output voltage range over which the digital input is translated into an analog signal. For both ADCs and DACs, low-noise design techniques, shielding, buffering, and filtering strategies are often employed to protect the integrity of the reference voltage.

SUMMARY

[0006] In accordance with an embodiment, a circuit includes an amplifier comprising a first input coupled to a reference voltage node, a first buffer having an input coupled to an output of the amplifier, a second buffer having an input coupled to the output of the amplifier, and a capacitor. A method of operating the circuit includes during a first phase of operation: forming a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sampling a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, forming a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

[0007] In accordance with another embodiment, a circuit includes: an amplifier comprising a first input coupled to a reference voltage node; a first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; and a controller configured to: during a first phase of operation: form a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sample a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, form a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

[0008] In accordance with a further embodiment, a circuit includes: an amplifier having a first input coupled to a reference voltage node and a second input; a first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; a first switch coupled between an output of the first buffer and the second input of the amplifier; a second switch coupled between the first input of the amplifier and a first terminal of

the capacitor; a third switch coupled between an output of the second buffer and a second terminal of the capacitor; a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor; and a fifth switch coupled between the output of the first buffer and the first terminal of the amplifier.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0010] FIG. 1 illustrates a reference buffer circuit according to an embodiment;

[0011] FIG. 2 illustrates a schematic of circuitry used to implement portions of the reference buffer circuit of FIG. 1;

[0012] FIGS. 3A and 3B are schematics of a successive approximation analog-to-digital converter that includes the reference buffer circuit of FIG. 1; and

[0013] FIG. 4 is a block diagram of a method according to an embodiment.

[0014] Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale. To more clearly illustrate certain embodiments, a letter indicating variations of the same structure, material, or process step may follow a figure number.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0015] The making and using of the presently preferred embodiments are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

[0016] In an embodiment, a reference voltage buffer circuit includes an amplifier with its output coupled to both an output buffer and a replica buffer, with the replica buffer intended for connection to an external load. The system operates in two distinct phases. In the first phase, the circuit captures the offset voltage that exists between the differential amplifier's first input terminal and the output of the replica buffer, storing this value on a dedicated capacitor. Subsequently, in the second phase, an offset correction feedback loop is established that includes the amplifier, the capacitor, and the output buffer. The purpose of this loop is to adjust the voltage at the second input terminal of the amplifier by subtracting the previously stored offset voltage from it. Concurrently, during this phase, the replica buffer actively delivers a replica of the voltage present at the differential amplifier's first input terminal.

[0017] Because the replica buffer operates in an open loop configuration during the second phase, the bandwidth of the replica buffer is not limited by the loop bandwidth of the offset correction feedback loop. Thus, the replica buffer can advantageously provide a buffered voltage that is stable over a large bandwidth with high phase margins even when driving capacitive loads in some embodiments. Moreover, even though the replica buffer is not a part of the offset correction feedback loop during the second phase, the replica buffer is advantageously able to provide a very low offset signal due to its offset being stored on the capacitor.

[0018] In some embodiments, the output buffer and/or amplifier may be biased at a lower current than the replica buffer in order to save power. In some embodiments, the reference voltage buffer is configured to buffer a reference voltage for an analog-to-digital converter.

[0019] FIG. 1 illustrates reference voltage buffer circuit 100 according to an embodiment of the

present invention. As shown, reference voltage buffer circuit **100** includes an amplifier **102**, an output buffer **104**, a replica buffer **106**, a set of switches **S1**, **S2**, **S3**, **S4**, and **S5**, and a capacitor **C**. The amplifier **102** has a first input coupled to a reference voltage node, and the output buffer **104** and the replica buffer **106** each have an input coupled to an output of the amplifier **102**. The replica buffer **106** provides the voltage $V_{\text{sub.rep}}$ to a load, while the output buffer **104** generates a voltage $V_{\text{sub.out}}$. The switches **S1** through **S5** are arranged to facilitate a dynamic offset cancellation process during different phases of operation, as indicated by the timing diagrams for phases $\phi_{\text{sub.1}}$ and $\phi_{\text{sub.2}}$. The capacitor **C** is used to sample and store the offset error during the first phase $\phi_{\text{sub.1}}$ of operation and to subtract this error from the input during the second phase $\phi_{\text{sub.2}}$ of operation, thereby correcting for the intrinsic offset of the reference voltage buffer circuit **100**.

[0020] FIG. **1** also illustrates a timing diagram showing the relationship between the first phase $\phi_{\text{sub.1}}$ and second phase $\phi_{\text{sub.2}}$ of operation. In some embodiments the first phase $\phi_{\text{sub.1}}$ and second phase $\phi_{\text{sub.2}}$ are controlled using non-overlapping clock signals to prevent loss of stored charge from capacitor **C**. These clock signals may be generated, for example, using a controller (not shown) configured to generate the non-overlapping clock signals. The controller may be implemented using, for example, non-overlapping clock generation circuits known in the art, using, for example, cross-coupled logic gates. The controller may also be implemented using other types of hardware and/or a combination of hardware, software or firmware. For example, in some embodiments, the controller may be implemented using a processor configured to execute software or firmware instructions stored in memory.

[0021] During the first phase $\phi_{\text{sub.1}}$ of operation switch **S1** is closed (also referred to as activated), which couples the output of output buffer **104** to the negative input (also referred to as the second input) of amplifier **102**. This essentially places amplifier **102** in a unity gain configuration. Switches **S2** and **S3** are also closed during the first phase of operation, which couples capacitor **C** between the output of replica buffer **106** and the positive input (also referred to as the first input) of amplifier **102** at reference voltage input node V_{ref} . Hence, during the first phase $\phi_{\text{sub.1}}$ of operation, a closed loop is formed that includes amplifier **102** and output buffer **104** (also referred to as a first buffer), and excludes capacitor **C** and replica buffer **106** (also referred to as a second buffer).

[0022] During the first phase of operation, the output voltage $v_{\text{sub.rep}}$ of replica buffer **106** is:

$$[00001] v_{\text{rep}} = v_{\text{ref}} + v_{\text{off}} + v_{\text{mis}},$$

where $v_{\text{sub.ref}}$ is the voltage at the positive input of amplifier **102**, v_{off} is the intrinsic offset of amplifier **102**, and $v_{\text{sub.mis}}$ is the mismatch between output buffer **104** and replica buffer **106**.

Accordingly, the voltage across capacitor **C** is:

$$[00002] v_{\text{cap}} = v_{\text{ref}} - v_{\text{rep}}.$$

[0023] During the second phase $\phi_{\text{sub.2}}$ of operation, switches **S4** and **S5** are closed and switches **S1**, **S2** and **S3** are opened (also referred to as deactivated). This essentially reverses the polarity of capacitor **C**, and forms a closed loop that includes amplifier **102**, output buffer **104** and capacitor **C** connected between the output of output buffer **104** and the negative input of amplifier **102**. Replica buffer continues to provide voltage $v_{\text{sub.rep}}$ to load **108**, but remains outside of the closed loop. Hence, during the second phase $\phi_{\text{sub.2}}$ of operation a closed loop is formed that includes amplifier **102**, output buffer **104**, and capacitor **C** with voltage $v_{\text{sub.cap}}$ stored thereon, and excludes replica buffer **106**. While capacitor **C** is shown coupled between the output of output amplifier **104** and the input of amplifier **102** in FIG. **1**, it should be understood that in alternative embodiments, capacitor **C** may be coupled in other positions within the closed loop. For example, capacitor **C** may be coupled between the output of amplifier **102** and the input of output **104** during the second phase $\phi_{\text{sub.2}}$ of operation.

[0024] During the second phase the replica output is:

$$\begin{aligned}
V_{\text{rep2}} &= V_{\text{ref}} + V_{\text{off}} + V_{\text{mis}} + V_{\text{cap}} \\
&= V_{\text{ref}} + V_{\text{off}} + V_{\text{mis}} + V_{\text{ref}} - V_{\text{rep1}} \\
[00003] \quad &= V_{\text{ref}} + V_{\text{off}} + V_{\text{mis}} + V_{\text{ref}} - V_{\text{ref}} - V_{\text{off}} - V_{\text{mis}} \\
&= V_{\text{ref}} .
\end{aligned}$$

As seen by the expressions above, embodiments of the present invention may provide compensation for both the intrinsic offset $v_{\text{sub.off}}$ of amplifier **102**, and the mismatch $v_{\text{sub.off}}$ between output buffer **104** and replica buffer **106**.

[0025] In accordance with an embodiment, each switch **S1**, **S2**, **S3**, **S4** and **S5** may be implemented, for example, using one or more transistors, such as NMOS or PMOS transistors. In some embodiments, one or more of switches **S1**, **S2**, **S3**, **S4** and **S5** may be implemented using CMOS transmission gates, which are formed by coupling one or more NMOS transistors in parallel with one or more PMOS transistors. In such embodiments, the gate of the transistor acts as the control node of the switch, while the source-drain load path of the transistor acts as the switchable conductive path.

[0026] Amplifier **102** may be implemented using a combination of transistors arranged to form an operational amplifier (OPAMP), a transconductance amplifier, or other type of suitable amplifier. In some embodiments, amplifier **102** has a high dc gain in order to reduce steady-state dc error when amplifier **102** and output buffer are arranged in a closed-loop configuration. Output buffer **104** and replica buffer **106** may be implemented using unity gain buffer circuits known in the art, such as emitter follower circuits. In some embodiments, replica buffer **106** may be biased with a higher current than output buffer **104** in order to provide a higher drive strength to drive load **108**. In some embodiments, load **108** may be a capacitive load, such as a capacitive array-based DAC as discussed herein.

[0027] Replica buffer **106** may use the same circuit topology as output buffer **104**, and may use physically larger devices than output buffer **104**. In some embodiments, output buffer **104** and replica buffer **106** are implemented using the same size unit devices, where replica buffer **106** includes a multiple of n unit devices with respect to output buffer **104** in order to enhance matching between output buffer **104** and replica buffer **106**. In such embodiments, replica buffer **106** may be biased with a current that is a multiple of the same scaling factor n applied to the relative device sizes between output buffer **104** and replica buffer **106**.

[0028] In various embodiments, during the second phase $\phi_{\text{sub.2}}$ of operation, the closed loop bandwidth of amplifier **102** and output buffer **104** has a lower bandwidth than replica buffer **106**. This advantageously permits the loop including amplifier **102** and output buffer **104** to provide accurate offset cancelation at the time that replica buffer provides a fast, high bandwidth reference voltage to load **108**.

[0029] In various embodiments, the components of buffer circuit **100** may be disposed on a single monolithic semiconductor integrated circuit, such as a single semiconductor substrate, and/or on the same monolithic semiconductor integrated circuit as other disclosed system components. The semiconductor substrate may be a silicon substrate. Buffer circuit **100** may be fabricated using one of a variety of different semiconductor processes, such as a CMOS process, BiCMOS process, Bipolar process, or other type of process.

[0030] FIG. 2 illustrates an example implementation of an amplifier **202**, output buffer **204** and replica buffer **206** that could be used to respectively implement amplifier **102**, output buffer **104** and replica buffer **106** shown in FIG. 1. Bias generator **201** is configured to provide bias voltages to amplifier **202**, output buffer **204** and replica buffer **206**.

[0031] As shown, amplifier **202** is implemented as a folded cascode transconductance amplifier whose input stage includes NMOS transistors **MN1** and **MN2** that form a differential input stage with a tail current provided by NMOS transistor **MN3**. The drains of transistors **MN1** and **MN2** are respectively connected to the drains of PMOS current source transistors **MP1** and **MP2**, and

respectively connected to the sources of PMOS cascode transistors MP3 and MP4. The output stage also includes a cascoded current mirror made of NMOS transistors MN4, MN5, MN6 and MN7, where NMOS transistors MN6 and MN7 form a current mirror, and MN4 and MN5 serve as cascode transistors. The bias currents provided by transistors MN3, MP1 and MP2, and the cascode bias voltage provided to transistors MP3, MP4, MN4 and MN5 may be set via bias generator 201 using standard bias generation circuits and techniques known in the art.

[0032] The output of amplifier 202 is taken from the drains of PMOS transistor MP4 and NMOS transistor MN5 and may be loaded with capacitor C1. In some embodiments, the closed loop bandwidth of amplifier 202 and output buffer 204 when connected in a feedback configuration is dependent on the capacitance of capacitor C1. The capacitance of capacitor C1 may be set, for example, to ensure an adequate phase margin according to circuit design practice known in the art.

[0033] It should be appreciated that the topology of amplifier 202 is just one example of a number of different ways amplifier 202 could be implemented. For example, in an alternative embodiment, NMOS cascode transistors MN4 and MN5 could be removed or a telescopic cascode structure could be used instead of a folded cascode structure. In some embodiments, the structure of amplifier 202 could be inverted such that a PMOS-based input stage is used instead of an NMOS-based input. In such an embodiment, the cascode output stage could be modified such that the output of the PMOS-based input stage is coupled to NMOS current source transistors and NMOS cascode transistors instead of PMOS current source transistors MP1 and MP2 and PMOS cascode transistors MP3 and MP4.

[0034] FIG. 2 shows output buffer 204 and replica buffer 206 as being implemented using flipped voltage follower (FVF) circuits, which is a type of source follower amplifier. For example, output buffer 204 includes a PMOS source follower transistor MP20 that is coupled to the output of amplifier 202. The drain of PMOS source follower transistor MP20 is connected to the drain of NMOS current source transistor MN20. The drains of transistors MP20 and MN20 are connected to the gate of PMOS transistor MP22 that supplies current to PMOS source follower transistor MP20, which forms a local feedback loop that keeps the current through PMOS source follower transistor MP20 substantially constant, thereby causing output buffer 204 to substantially have a unity gain.

[0035] Replica buffer 206 has the same circuit topology as output buffer 204 and includes PMOS source follower transistor MP10, current source transistor MN10 and PMOS transistor MP12 that supplies current to the source of PMOS source follower transistor MP10. As shown, each transistor of replica buffer 206 has an aspect ratio that is n times larger than the aspect ratio of each respective transistor in output buffer 204. In some embodiments, each transistor in output buffer 204 and replica buffer 206 use the same unit cells, such that replica buffer 206 uses n times the number of unit cells for each device than output buffer 204. This also causes the bias current of replica buffer 206 to be n times larger than the bias current of output buffer 204 since NMOS current source transistor MN10 of replica buffer 206 has an aspect ratio that is n times larger than the aspect ratio of NMOS current source transistor MN20 in output buffer 204. As shown, the gates of current source transistors MN10 and MN20 are connected to bias generator 201.

[0036] In alternative embodiments, output buffer 204 and replica buffer 206 may be implemented using other buffer or amplifier circuits known in the art. For example, output buffer 204 and replica buffer 206 may be implemented using simple source follower circuits instead of FVF circuits.

[0037] The arrangement of these components may form a system designed to provide a stable reference voltage with reduced offset error for analog-to-digital converters. FIG. 3A illustrates an example successive approximation register analog-to-digital converter SAR-ADC 300 that utilizes a reference voltage buffer circuit 100 according to an embodiment. As shown, SAR ADC 300 includes a digital-to-analog converter (DAC) 302, a comparator 304, and a successive approximation register (SAR) 306. Buffer circuit 100, as described according to embodiments herein, provides a buffered reference voltage $V_{\text{sub.REFB}}$ to DAC 302 based derived from reference voltage V_{REF} provided by voltage reference circuit 310. For example, the output of

replica buffer **106** of buffer circuit **100** shown in FIG. **1** may drive the capacitive array of DAC **302** during the second phase $\phi_{\text{sub.2}}$ of operation. In some embodiments, a lowpass filter **312** may be coupled between voltage reference circuit **310** and buffer circuit **100** to filter noise and transients prior to buffering.

[0038] In various embodiments, voltage reference circuit **310** may be implemented, for example, using voltage reference circuits known in the art, such as a bandgap voltage generator. Lowpass filter **312** may be implemented, for example, using a simple passive RC filter. Alternatively, lowpass filter **312** may be a higher order filter and/or an active filter.

[0039] As shown, DAC **302** is implemented as a capacitor array (also referred to as a capacitive array) DAC that includes a binary-weighted array of capacitors with capacitance values that are multiples of a base unit capacitor: C , $2C$, $4C$, . . . up to NC . The top plates of the capacitors of the capacitor array are coupled to the negative input of comparator **304**, while the bottom plates of the capacitors are coupled to individual switches in switch array **308**. The states of switch array is controlled by the DATA signal produced by SAR **306**.

[0040] Operation of SAR ADC **300** may proceed according to known capacitor array-based SAR methods. For example, a sampling phase is initiated in which input voltage V_{IN} is applied to the bottom plates of the capacitor array via switches of switch array **308** while the top plate of the capacitor array is grounded by closing switch **320**. Next, SAR-ADC enters a successive approximation phase in which the contents of successive approximation register (SAR) **306** is initialized, for example, with all ones, switch **320** is opened, and the bottom plates of the capacitor array are selectively connected to node $V_{\text{sub.REFB}}$ and node GND over a number of successive approximation cycles. For example, during a first cycle, switch **322** is closed, which connects the rightmost capacitor C to node GND, while the remaining switches of switch array **308** apply node $V_{\text{sub.REFB}}$ to the bottom plates of the remaining capacitors. This causes the voltage of the top plates of the capacitor array to assume a voltage that represents the difference between an analog value representative of the digital value DATA and the input voltage V_{IN} . Next, comparator **304** determines whether the voltage of the top plate of the capacitor is greater or less than ground voltage GND. Next, the most significant bit of SAR **306** is set to the output value of comparator **304**, which causes a corresponding change to the switch states of switch array **308**. This process continues successively through all bits from the MSB down to the least significant bit (LSB), effectively approximating the input voltage V_{IN} . Each step refines the approximation until after N cycles (for an N -bit ADC), SAR **306** contains a digital code that corresponds to V_{IN} within the resolution limits of the ADC. The operation of SAR ADC **300** may be synchronized by clock generator **330**, control logic and/or a controller (not shown) that manages timing and control signals for switches and for data handling between DAC **302**, comparator **304**, and SAR **306**.

[0041] It should be understood that the illustrated structure of DAC **302** is just one example of many possible structures that could be used to implement the function of DAC **302**. In alternative embodiments, DAC **302** may be implemented using two capacitor arrays and/or may be operated in a differential manner. In some embodiments, a different voltage besides ground voltage may be applied to the bottom plates of the capacitor array and to comparator **304**, for example, in order to facilitate operation with a single-ended power supply. In alternative embodiments, DAC **302** may be implemented using other DAC architectures known in the art including, but not limited to resistor-string DACs, R-2R ladder DACs, and current-steering DACs. In embodiments that do not perform bottom-plate sampling of input voltage V_{IN} , SAR-ADC may include a separate sample-and-hold circuit to sample input voltage V_{IN} and/or a subtraction circuit configured to form a difference between input voltage V_{IN} and the analog output value of DAC **302**.

[0042] In further alternative embodiments, buffer circuit **100** may be used to provide a reference voltage to a non-SAR ADC including, but not limited to, a flash ADC, a pipelined ADC, a sigma-delta ADC, and the like, as well as non-ADC circuits such as DACs, bias generators, and other types of analog circuits.

[0043] Clock generator **330** provides a low frequency clock signal PCLK for controlling the operation of embodiment buffer circuit **100**, and a higher frequency clock signal CLK for clocking SAR **306**, which controls the input of DAC **302** and digital output DATA. Representative clock signals PCLK and CLK are shown in the timing diagram of FIG. **3B**. During first phase Φ .sub.1 of PCLK the offset of buffer amplifier is stored in a capacitor as described above with respect to FIG. **1**. During the second phase Φ .sub.2 of PCLK, the replica buffer of buffer circuit **100** provides a low-offset buffered reference voltage V.sub.REFB to DAC **302**. At the same time, DAC **302**, comparator **304** and SAR **306** sample and successively approximate input voltage VIN according to the higher frequency clock signal CLK.

[0044] FIG. **4** illustrates a method **400** of operating a circuit according to an embodiment. The circuit includes an amplifier having a first input coupled to a reference voltage node, a first buffer having an input coupled to an output of the amplifier, a second buffer having an input coupled to the output of the amplifier, and a capacitor. The circuit may be buffer circuit **100** shown in FIG. **1** and described above. The amplifier, first buffer and second buffer may respectively correspond, for example, to amplifier **102**, output buffer **104**, and replica buffer **106** as shown and described above with respect to FIG. **1**. The capacitor may correspond to capacitor C shown in FIG. **1**.

[0045] As shown, method **400** includes two phases of operation: phase **1** (step **402**) and phase **2** (step **410**). During the first phase (step **402**), a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer is formed (step **404**), and the voltage difference between an output of the second buffer and the reference voltage node across the capacitor is sampled by the capacitor (step **406**). In some embodiments, this voltage difference represents the offset voltage that exists between the reference voltage node and the output of the second buffer. The first phase of operation may be implemented, for example, by closing switches S1, S2 and S3 and opening switches S4 and S5 as described above with respect to FIG. **1**.

Operation continues in phase **1** (step **402**) as long as the phase signal @, is asserted (step **408**).

[0046] During the second phase (step **410**), a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer is formed (step **412**). This second phase of operation may be implemented, for example, by opening switches S1, S2 and S3 and closing switches S4 and S5 as described above with respect to FIG. **1**. In the second phase of operation, a load may be driven using the output of the second buffer. Operation continues in phase **2** (step **410**) as long as the phase signal Φ .sub.2 is asserted (step **414**).

[0047] Embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

[0048] Example 1. A method of operating a circuit including an amplifier including a first input coupled to a reference voltage node, a first buffer having an input coupled to an output of the amplifier, a second buffer having an input coupled to the output of the amplifier, and a capacitor, the method includes: during a first phase of operation: forming a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sampling a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, forming a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

[0049] Example 2. The method of example 1, where the output of the second buffer is coupled to a capacitive load.

[0050] Example 3. The method of example 1 or 2, where forming the first closed loop includes closing a first switch coupled between an output of the first buffer and a second input of the amplifier; sampling the voltage difference includes closing a second switch coupled between the reference voltage node and a first terminal of the capacitor, and closing a third switch coupled between the output of the second buffer and a second terminal of the capacitor; and forming the

second closed loop includes closing a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor, and closing a fifth switch coupled between the output of the first buffer and the first terminal of the capacitor.

[0051] Example 4. The method of any one of examples 1 to 3, further including: opening the fourth switch and the fifth switch during the first phase of operation; and opening the first switch, the second switch and the third switch during the second phase of operation.

[0052] Example 5. The method of example 4, further including driving, by the output of the second buffer, a capacitive array of an analog-to-digital converter during the second phase of operation.

[0053] Example 6. The method of any one of examples 1 to 5, where, during the second phase of operation, the second closed loop has a lower bandwidth than the second buffer.

[0054] Example 7. A circuit, including: an amplifier including a first input coupled to a reference voltage node; a first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; and a controller configured to: during a first phase of operation: form a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sample a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, form a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

[0055] Example 8. The circuit of example 7, further including: a first switch coupled between an output of the first buffer and a second input of the amplifier; a second switch coupled between the reference voltage node and a first terminal of the capacitor; a third switch coupled between the output of the second buffer and a second terminal of the capacitor; a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor; and a fifth switch coupled between the output of the first buffer and the first terminal of the amplifier.

[0056] Example 9. The circuit of example 8, where the controller is configured to: during the first phase of operation: close the first switch, the second switch and the third switch, and open the fourth switch and the fifth switch; and during the second phase of operation: open the first switch, the second switch and the third switch, and close the fourth switch and the fifth switch.

[0057] Example 10. The circuit of any one of examples 7 to 9, where the first input of the amplifier includes a positive input, and the second input of the amplifier includes a negative input.

[0058] Example 11. The circuit of any one of examples 7 to 10, where the first buffer is configured to be biased with a lower current than the second buffer.

[0059] Example 12. The circuit of any one of examples 7 to 11, where the first buffer includes a first source follower amplifier; and the second buffer includes a second source follower amplifier.

[0060] Example 13. The circuit of example 12, where the first source follower amplifier includes a first flipped voltage follower (FVF) circuit; and the second source follower amplifier includes a second FVF circuit.

[0061] Example 14. The circuit of any one of examples 7 to 13, where, during the second phase of operation, the second closed loop has a lower bandwidth than the second buffer.

[0062] Example 15. The circuit of any one of examples 7 to 14, where during the first phase of operation, an intrinsic offset voltage of the amplifier is stored on the capacitor; and during the second phase of operation, the intrinsic offset voltage stored on the capacitor is subtracted from a second input of the amplifier.

[0063] Example 16. A circuit including: an amplifier having a first input coupled to a reference voltage node and a second input; a first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; a first switch coupled between an output of the first buffer and the second input of the amplifier; a second switch coupled between the first input of the amplifier and a first terminal of the capacitor; a third switch coupled between an output of the second buffer and a second terminal of the capacitor; a fourth switch coupled between the second input of the amplifier and the second terminal of the

capacitor; and a fifth switch coupled between the output of the first buffer and the first terminal of the amplifier.

[0064] Example 17. The circuit of example 16, further including a controller coupled to control nodes of the first switch, the second switch, the third switch, the fourth switch and the fifth switch, the controller configured to: during a first phase of operation: activate the first switch, the second switch and the third switch, and deactivate the fourth switch and the fifth switch; and during a second phase of operation: deactivate the first switch, the second switch and the third switch, and activate the fourth switch and the fifth switch.

[0065] Example 18. The circuit of example 16 or 17, where the amplifier includes a folded cascode transconductance amplifier; the first buffer includes a first source follower amplifier; and the second buffer includes a second source follower amplifier.

[0066] Example 19. The circuit of example 18, where the first source follower amplifier includes a first flipped voltage follower (FVF) circuit; and the second source follower amplifier includes a second FVF circuit.

[0067] Example 20. The circuit of example 18 or 19, where the second buffer is configured to have a higher bias current than the first buffer.

[0068] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of operating a circuit comprising an amplifier comprising a first input coupled to a reference voltage node, a first buffer having an input coupled to an output of the amplifier, a second buffer having an input coupled to the output of the amplifier, and a capacitor, the method comprising: during a first phase of operation: forming a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sampling a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, forming a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.
2. The method of claim 1, wherein the output of the second buffer is coupled to a capacitive load.
3. The method of claim 1, wherein: forming the first closed loop comprises closing a first switch coupled between an output of the first buffer and a second input of the amplifier; sampling the voltage difference comprises closing a second switch coupled between the reference voltage node and a first terminal of the capacitor, and closing a third switch coupled between the output of the second buffer and a second terminal of the capacitor; and forming the second closed loop comprises closing a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor, and closing a fifth switch coupled between the output of the first buffer and the first terminal of the capacitor.
4. The method of claim 3, further comprising: opening the fourth switch and the fifth switch during the first phase of operation; and opening the first switch, the second switch and the third switch during the second phase of operation.
5. The method of claim 1, further comprising driving, by the output of the second buffer, a capacitive array of an analog-to-digital converter during the second phase of operation.
6. The method of claim 1, wherein, during the second phase of operation, the second closed loop has a lower bandwidth than the second buffer.
7. A circuit, comprising: an amplifier comprising a first input coupled to a reference voltage node; a

first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; and a controller configured to: during a first phase of operation: form a first closed loop including the amplifier and the first buffer and excluding the capacitor and the second buffer, and sample a voltage difference between an output of the second buffer and the reference voltage node across the capacitor; and during a second phase of operation, form a second closed loop including the amplifier, the first buffer, and the capacitor with the sampled voltage difference stored thereon, and excluding the second buffer.

8. The circuit of claim 7, further comprising: a first switch coupled between an output of the first buffer and a second input of the amplifier; a second switch coupled between the reference voltage node and a first terminal of the capacitor; a third switch coupled between the output of the second buffer and a second terminal of the capacitor; a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor; and a fifth switch coupled between the output of the first buffer and the first terminal of the amplifier.

9. The circuit of claim 8, wherein the controller is configured to: during the first phase of operation: close the first switch, the second switch and the third switch, and open the fourth switch and the fifth switch; and during the second phase of operation: open the first switch, the second switch and the third switch, and close the fourth switch and the fifth switch.

10. The circuit of claim 8, wherein the first input of the amplifier comprises a positive input, and the second input of the amplifier comprises a negative input.

11. The circuit of claim 7, wherein the first buffer is configured to be biased with a lower current than the second buffer.

12. The circuit of claim 7, wherein: the first buffer comprises a first source follower amplifier; and the second buffer comprises a second source follower amplifier.

13. The circuit of claim 12, wherein: the first source follower amplifier comprises a first flipped voltage follower (FVF) circuit; and the second source follower amplifier comprises a second FVF circuit.

14. The circuit of claim 7, wherein, during the second phase of operation, the second closed loop has a lower bandwidth than the second buffer.

15. The circuit of claim 7, wherein: during the first phase of operation, an intrinsic offset voltage of the amplifier is stored on the capacitor; and during the second phase of operation, the intrinsic offset voltage stored on the capacitor is subtracted from a second input of the amplifier.

16. A circuit comprising: an amplifier having a first input coupled to a reference voltage node and a second input; a first buffer having an input coupled to an output of the amplifier; a second buffer having an input coupled to the output of the amplifier; a capacitor; a first switch coupled between an output of the first buffer and the second input of the amplifier; a second switch coupled between the first input of the amplifier and a first terminal of the capacitor; a third switch coupled between an output of the second buffer and a second terminal of the capacitor; a fourth switch coupled between the second input of the amplifier and the second terminal of the capacitor; and a fifth switch coupled between the output of the first buffer and the first terminal of the amplifier.

17. The circuit of claim 16, further comprising a controller coupled to control nodes of the first switch, the second switch, the third switch, the fourth switch and the fifth switch, the controller configured to: during a first phase of operation: activate the first switch, the second switch and the third switch, and deactivate the fourth switch and the fifth switch; and during a second phase of operation: deactivate the first switch, the second switch and the third switch, and activate the fourth switch and the fifth switch.

18. The circuit of claim 16, wherein: the amplifier comprises a folded cascode transconductance amplifier; the first buffer comprises a first source follower amplifier; and the second buffer comprises a second source follower amplifier.

19. The circuit of claim 18, wherein: the first source follower amplifier comprises a first flipped voltage follower (FVF) circuit; and the second source follower amplifier comprises a second FVF

circuit.

20. The circuit of claim 16, wherein the second buffer is configured to have a higher bias current than the first buffer.
